



# SPECIFICATION FOR CTP MODULE

MODULE NO: YB-TG720720S01A-C-A0

Doc.Version:00

Customer Approval:

|                                 |                                 |
|---------------------------------|---------------------------------|
| <input type="checkbox"/> Accept | <input type="checkbox"/> Reject |
|---------------------------------|---------------------------------|

| YEEBO    | NAME                | SIGNATURE | DATE       |
|----------|---------------------|-----------|------------|
| Prepare  | Electronic Engineer | 赵春风       | 2022-08-30 |
| Check    | Mechanical Engineer | 李毅斌       | 2022-08-30 |
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| Approval |                     | 孙五南       | 2022-08-31 |

■ APPROVAL FOR SPECIFICATIONS ONLY

☐ APPROVAL FOR SPECIFICATIONS AND SAMPLE

WIMRD005-02-D

## 1. Revision History

[illegible]

**2. Table of Contents:**

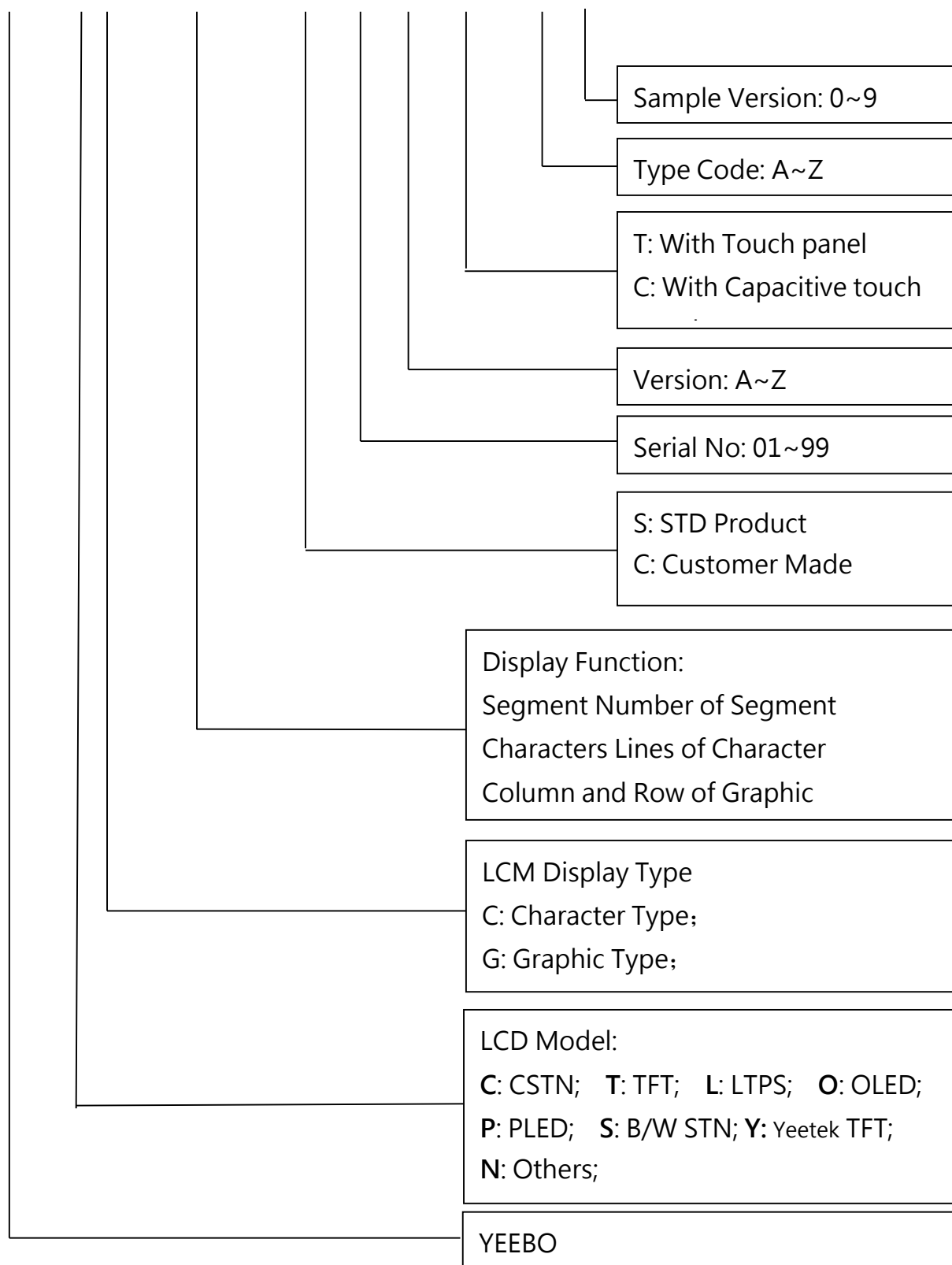
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### 3. Module Numbering System:

(Example)

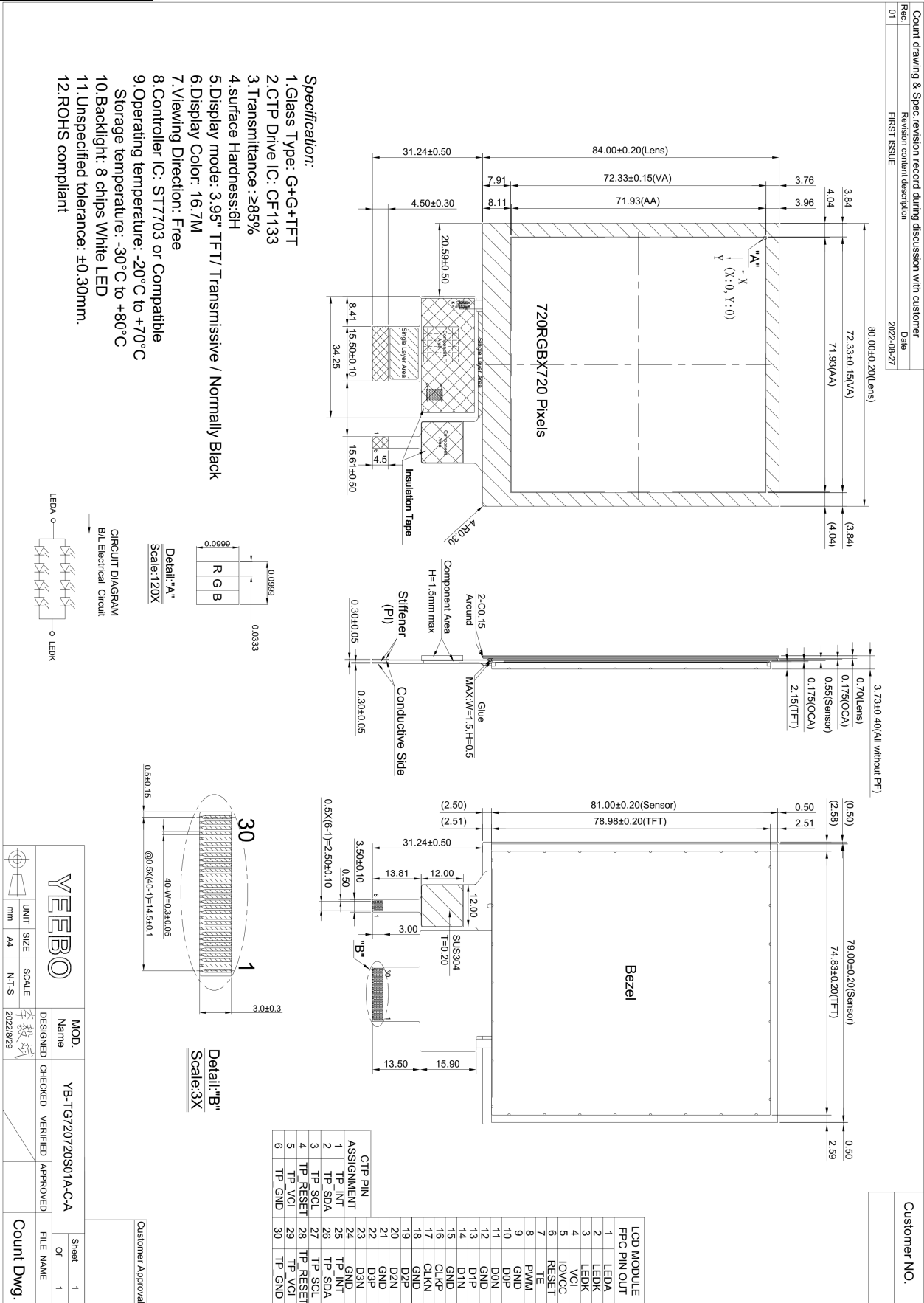
**YB- TG 720720 S 01 A-C – A 0**



**4. General Specification:**

| ITEM                       | CONTENTS                            |
|----------------------------|-------------------------------------|
| Module Size                | 80.00 (W) * 84.00 (H) * 3.73(T) mm  |
| Module Size (with FPC)     | 80.00 (W) * 115.25 (H) * 3.73(T) mm |
| Display Size (Diagonal)    | 3.95 inch                           |
| Display Format             | 720(RGB)*720 Pixels                 |
| Active Area                | 71.928(H) × 71.928(V)               |
| View Area                  | 72.33(H) × 72.33(V)                 |
| Pixel Pitch                | 0.0999(H) mm*0.0999 (V) mm          |
| LCD Type                   | TFT (16.7M)/ Transmissive / NB      |
| The Best Viewing Direction | FREE                                |
| Controller IC              | ST7703                              |
| CTP IC                     | CF1133                              |
| Weight                     | TBD                                 |

## 5. Drawing:



## 6. Electrical Characteristics

### 6-1 Absolute Maximum Ratings

#### 6-1-1 Absolute Maximum Ratings (TFT)

(Ta=25°C VSS=0V)

| Item                       | Symbol           | Min. | Type | Max.                  | Unit | Remark |
|----------------------------|------------------|------|------|-----------------------|------|--------|
| Power Supply Voltage       | V <sub>CI</sub>  | -0.3 | -    | 6.6                   | V    |        |
| Logic Input Voltage Range  | V <sub>IH</sub>  | -0.3 | -    | V <sub>DDI</sub> +0.3 | V    |        |
| Logic Output Voltage Range | V <sub>OH</sub>  | -0.3 | -    | V <sub>DDI</sub> +0.3 | V    |        |
| Operating Temperature      | T <sub>opr</sub> | -20  | -    | +70                   | °C   |        |
| Storage Temperature        | T <sub>stg</sub> | -30  | -    | +80                   | °C   |        |

Note1: Absolute maximum rating is the limit value beyond which the IC maybe broken.  
They do not assure operations.

#### 6-1-2 Absolute Maximum Ratings (CTP)

| Parameter                     | Symbol             | Min. | Max. | Unit |
|-------------------------------|--------------------|------|------|------|
| VDD                           | V <sub>VDD</sub>   | -0.3 | +6   | V    |
| IOVDD                         | V <sub>IOVDD</sub> | -0.3 | +6   | V    |
| Operating Ambient Temperature | T <sub>A</sub>     | -20  | +80  | °C   |
| Storage Temperature           | T <sub>S</sub>     | -40  | +125 | °C   |

**\*Note:** Stresses above those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. All the ranges are stress ratings only. Functional operation of this device at these or any other conditions above those indicated in the operational sections of this specification is not implied or intended. Exposed to the absolute maximum rating conditions for extended periods may affect device reliability.

**6-2 Operating Conditions (TFT)****(Ta=25°C )****6-2-1 Operating Conditions (TFT)****(Ta=25°C )**

| Item                 | Symbol          | Condition             | Min.                 | Type | Max.                 | Unit | Remark |
|----------------------|-----------------|-----------------------|----------------------|------|----------------------|------|--------|
| Power Supply Voltage | V <sub>CI</sub> | -                     | 2.5                  | 2.8  | 3.6                  | V    |        |
|                      | IOVCC           |                       | 1.65                 | 1.8  | 2.0                  | V    |        |
| IO Supply Voltage    | V <sub>IH</sub> | -                     | 0.7 V <sub>DDI</sub> | -    | V <sub>DDI</sub>     | V    |        |
|                      | V <sub>IL</sub> | -                     | V <sub>SS</sub>      | -    | 0.3 V <sub>DDI</sub> | V    |        |
|                      | V <sub>OH</sub> | -                     | 0.8 V <sub>DDI</sub> | -    | V <sub>DDI</sub>     | mA   |        |
|                      | V <sub>OL</sub> | -                     | V <sub>SS</sub>      | -    | 0.2 V <sub>DDI</sub> | V    |        |
| Power Supply Current | I <sub>DD</sub> | V <sub>CI</sub> =2.8V | -                    | 40   | 60                   | mA   |        |

**6-2-2 Operating Conditions (CTP)**Condition: VDD = IOVDD = 3.3V, T<sub>A</sub> = 25°C, unless be specified individually.

| Parameter              | Symbol             | Min.          | Typ. | Max.          | Unit | Condition                     |
|------------------------|--------------------|---------------|------|---------------|------|-------------------------------|
| VDD                    | V <sub>VDD</sub>   | 2.7           | -    | 3.6           | V    |                               |
| IOVDD                  | V <sub>IOVDD</sub> | 1.6           | -    | 3.6           | V    |                               |
| Operating Current      | I <sub>NML</sub>   | -             | -    | -             | mA   |                               |
| Idle Current           | I <sub>IDLE</sub>  | -             | -    | -             | mA   |                               |
| Power Down Current     | I <sub>PD</sub>    | -             | -    | -             | uA   |                               |
| Input High Voltage     | V <sub>IH</sub>    | 0.9*IO<br>VDD | -    | -             | V    |                               |
| Input Low Voltage      | V <sub>IL</sub>    | -             | -    | 0.1*IO<br>VDD | V    |                               |
| Input Pull Up Resistor | R <sub>PU</sub>    | 50            | -    | 60            | KOhm |                               |
| Output Driving Current | I <sub>DRV</sub>   | 6             | -    | -             | mA   | V <sub>OH</sub> = IOVDD x 0.8 |
| Output Sinking Current | I <sub>SINK</sub>  | 10            | -    | -             | mA   | V <sub>OL</sub> = IOVDD x 0.2 |
| Low Voltage Reset      | V <sub>LVR</sub>   | -             | -    | 2.3           | V    |                               |



## 6-3 AC Characteristics (TFT)

### 6-3-1 AC Characteristics (TFT)

#### DSI Interface Timing Characteristics

High Speed Mode

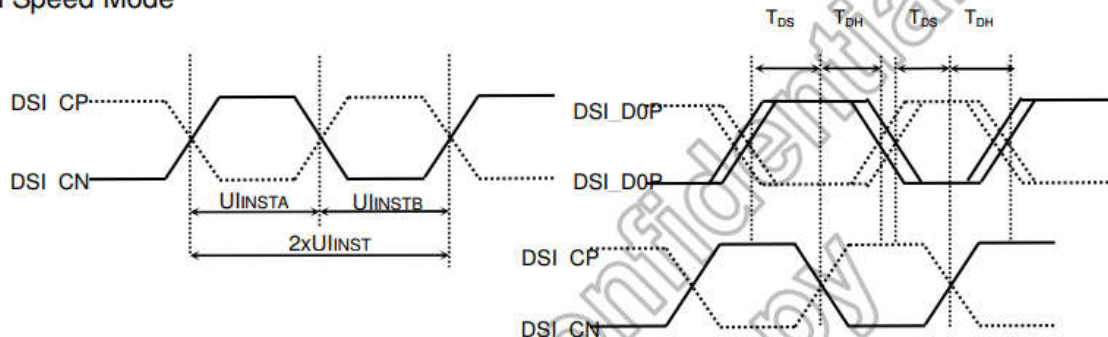


Figure 7.4: DSI clock timing Characteristics

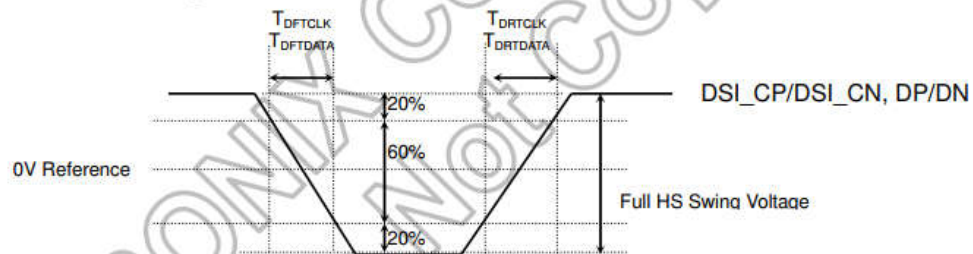


Figure 7.5: Rising and falling time on clock and data channel

(VSSA=0V, IOVCC=1.65V to 3.3V, VCI=2.5V to 3.3V,  $T_A = -30$  to  $70^\circ\text{C}$ )

| Signal            | Item                             | Symbol                       | Spec.     |      |         | Unit |
|-------------------|----------------------------------|------------------------------|-----------|------|---------|------|
|                   |                                  |                              | Min.      | Typ. | Max.    |      |
| DSI_CP/<br>DSI_CN | Double UI instantaneous          | $2xUI_{INST}$                | TBD       | -    | 25      | ns   |
|                   | UI instantaneous                 | $UI_{INSTA}$<br>$UI_{INSTB}$ | TBD       | -    | 12.5    | ns   |
| DP/DN             | Data to clock setup time         | $T_{DS}$                     | $0.15xUI$ | -    | -       | ps   |
|                   | Data to clock hold time          | $T_{DH}$                     | $0.15xUI$ | -    | -       | ps   |
| DSI_CP/<br>DSI_CN | Differential rise time for clock | $T_{DRTCLK}$                 | 150       | -    | $0.3UI$ | ps   |
|                   | Differential fall time for clock | $T_{DFTCLK}$                 | 150       | -    | $0.3UI$ | ps   |
| DP/DN             | Differential rise time for data  | $T_{DRTDATA}$                | 150       | -    | $0.3UI$ | ps   |
|                   | Differential fall time for data  | $T_{DFTDATA}$                | 150       | -    | $0.3UI$ | ps   |

Table 7.3: DSI High Speed Mode Characteristics

## Low Power Mode

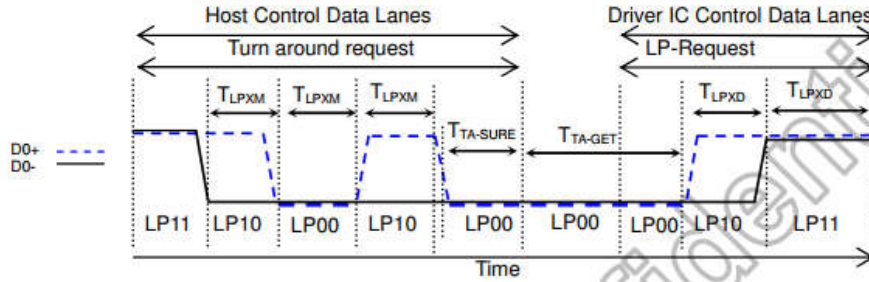


Figure 7.6: BTA from HOST to Display Module Timing

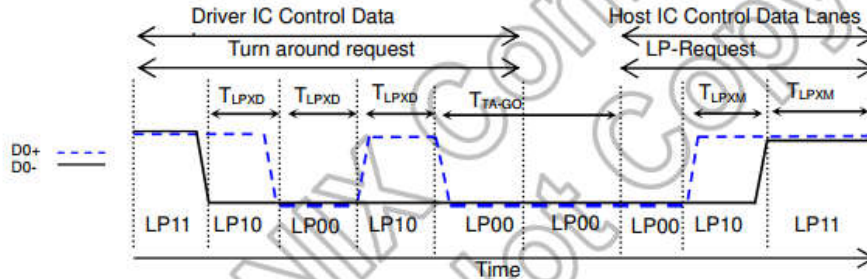


Figure 7.7: BTA from Display Module Timing to HOST

(VSSA=0V, IOVCC=1.65V to 3.3V, VCI=2.3V to 3.3V,  $T_A = -30$  to  $70^\circ\text{C}$ )

| Signal              | Item                                                    | Symbol        | Spec.               |      |                     | Unit |
|---------------------|---------------------------------------------------------|---------------|---------------------|------|---------------------|------|
|                     |                                                         |               | Min.                | Typ. | Max.                |      |
| DSI_D0P/<br>DSI_D0P | Length of LP-00/LP01/LP10/LP11<br>Host → Display module | $T_{LPXM}$    | 50                  | -    | -                   | ns   |
|                     | Length of LP-00/LP01/LP10/LP11<br>Display module → Host | $T_{LPXD}$    | 50                  | -    | -                   | ns   |
|                     | Time-out before the MPU start driver                    | $T_{TA-SURE}$ | $T_{LPXD}$          | -    | $2 \times T_{LPXD}$ | ns   |
|                     | Time to drive LP-00 by display module                   | $T_{TA-GET}$  | $5 \times T_{LPXD}$ | -    | -                   | ns   |
|                     | Time to drive LP-00 after turnaround request<br>Host    | $T_{TAGO}$    | $4 \times T_{LPXD}$ | -    | -                   | ns   |

Table 7.4: DSI Low Power Mode Characteristics

## Reset Timing:

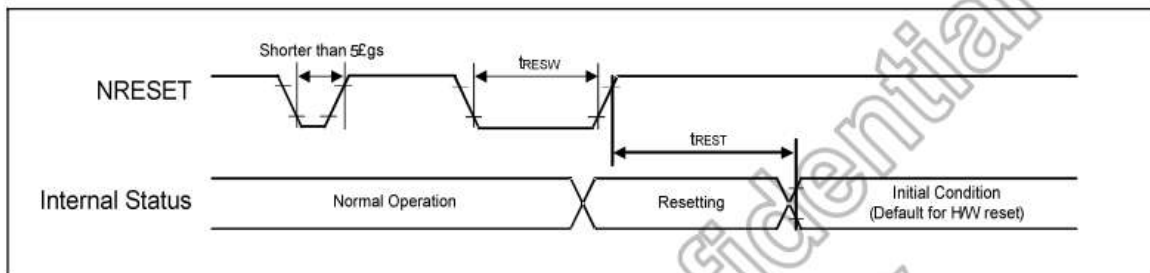
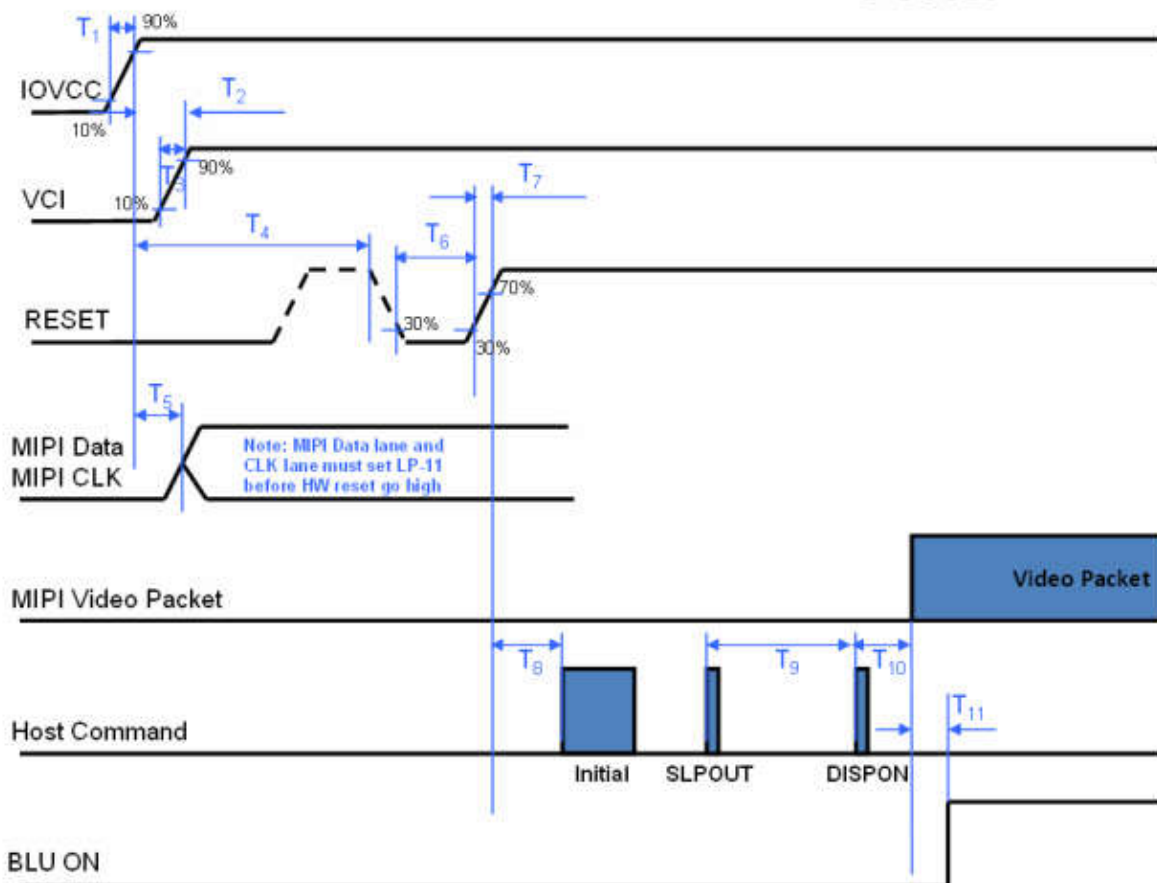


Figure 7.8: Reset input timing

| Symbol | Parameter                            | Related Pins | Spec. |      |      | Note                                  | Unit |
|--------|--------------------------------------|--------------|-------|------|------|---------------------------------------|------|
|        |                                      |              | Min.  | Typ. | Max. |                                       |      |
| tRESW  | Reset low pulse width <sup>(1)</sup> | NRESET       | 10    | -    | -    | -                                     | μs   |
| tREST  | Reset complete time <sup>(2)</sup>   | -            | 15    | -    | -    | When reset applied during SLPIN mode  | ms   |
|        |                                      | -            | 120   | -    | -    | When reset applied during SLPOUT mode | ms   |

Table 7.8: Reset Input Timing

**Power ON/OFF Sequence****Figure 8-6: DSI Power On Sequence of Power IC Mode**

|            | Min.     | Typ. | Max. | Unit |
|------------|----------|------|------|------|
| <b>T1</b>  | 0.01     | -    | 10   | ms   |
| <b>T2</b>  | No Limit |      |      | ms   |
| <b>T3</b>  | 0.01     | -    | 10   | ms   |
| <b>T4</b>  | 1        | -    | -    | ms   |
| <b>T5</b>  | 1        | -    | -    | ms   |
| <b>T6</b>  | 10       | -    | -    | us   |
| <b>T7</b>  | No Limit |      |      | ns   |
| <b>T8</b>  | 15       | -    | -    | ms   |
| <b>T9</b>  | 120      | -    | -    | ms   |
| <b>T10</b> | No Limit |      |      | ms   |
| <b>T11</b> | 100      | 150  | -    | ms   |

**Table 8-1: DSI Power On Timing of Power IC Mode**

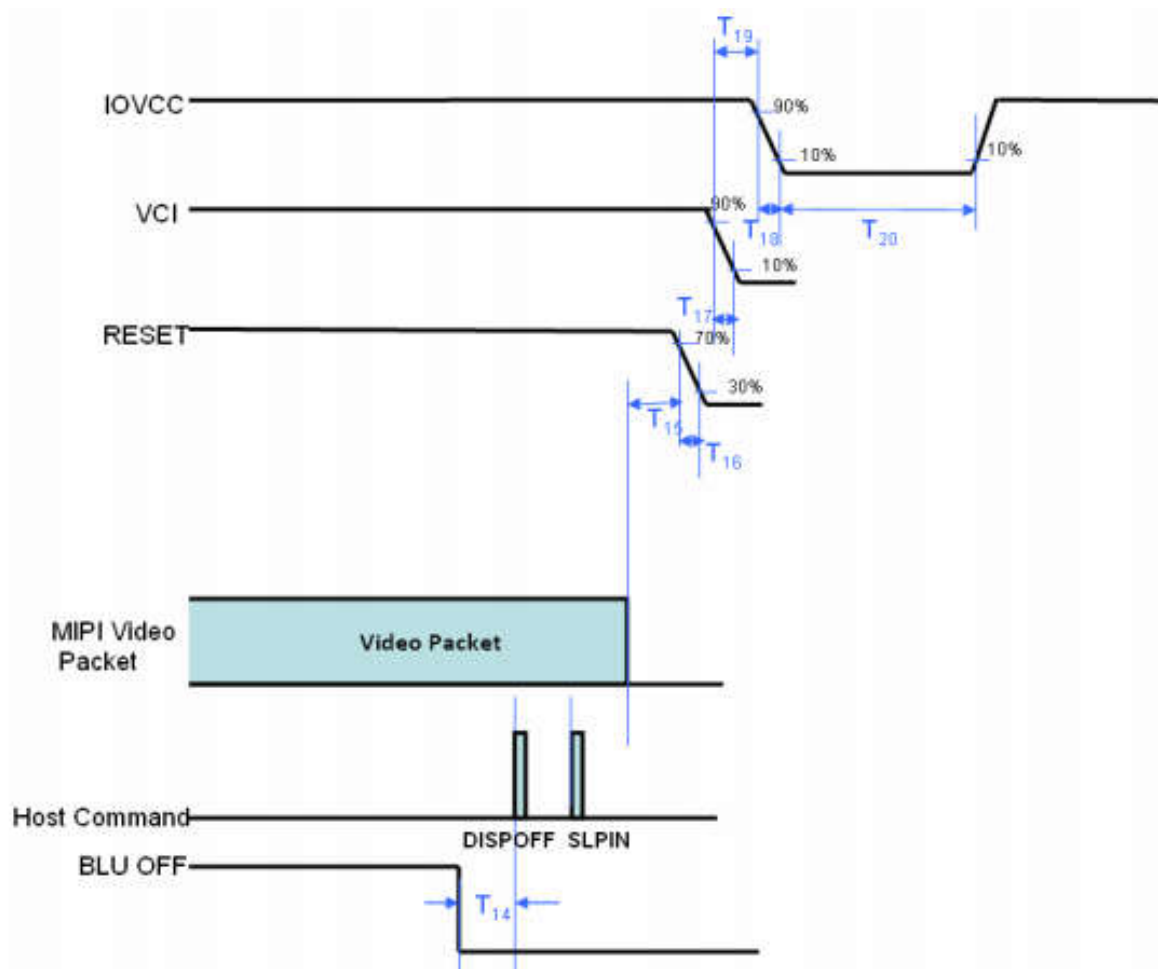


Figure 8-7: DSI Power Off Sequence of Power IC Mode

|     | Min.     | Typ. | Max. | Unit |
|-----|----------|------|------|------|
| T14 | 40       | 100  | -    | ms   |
| T15 | 10       | -    | -    | ms   |
| T16 | No Limit |      |      | ms   |
| T17 | No Limit |      |      | ms   |
| T18 | No Limit |      |      | ms   |
| T19 | No Limit |      |      | ms   |
| T20 | 500      |      |      | ms   |

Table 8-2: DSI Power Off Timing of Power IC Mode



## 6-3 AC Characteristics (CTP)

### 6-3-1 AC Characteristics (CTP)

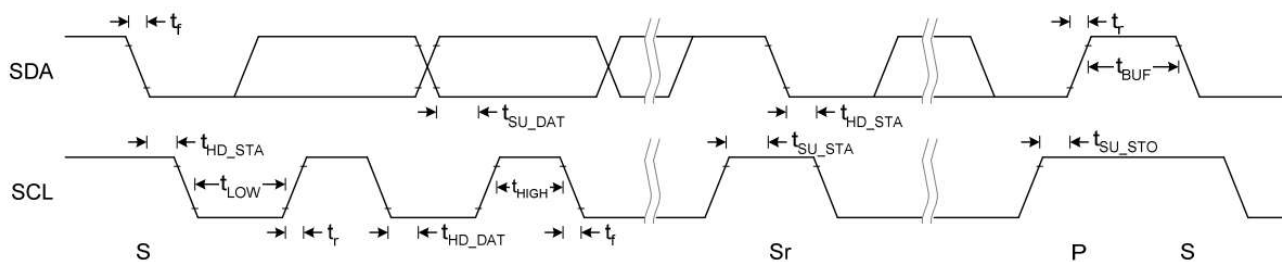


Figure 5-1 I2C Fast Mode Timing

Table 5-3 I2C Fast Mode Timing Characteristic

Conditions: VDD = 3.3V, GND = 0V, T<sub>A</sub> = 25°C

| Symbol              | Parameter                                                                                   | Rating |      |      | Unit |
|---------------------|---------------------------------------------------------------------------------------------|--------|------|------|------|
|                     |                                                                                             | Min.   | Typ. | Max. |      |
| f <sub>SCL</sub>    | SCL clock frequency                                                                         | 0      | -    | 400  | kHz  |
| t <sub>LOW</sub>    | Low period of the SCL clock                                                                 | 1.3    | -    | -    | us   |
| t <sub>HIGH</sub>   | High period of the SCL clock                                                                | 0.6    | -    | -    | us   |
| t <sub>f</sub>      | Signal falling time                                                                         | -      | -    | 300  | ns   |
| t <sub>r</sub>      | Signal rising time                                                                          | -      | -    | 300  | ns   |
| t <sub>SU_STA</sub> | Set up time for a repeated START condition                                                  | 0.6    | -    | -    | us   |
| t <sub>HD_STA</sub> | Hold time (repeated) START condition. After this period, the first clock pulse is generated | 0.6    | -    | -    | us   |
| t <sub>SU_DAT</sub> | Data set up time                                                                            | 100    | -    | -    | ns   |
| t <sub>HD_DAT</sub> | Data hold time                                                                              | 0      | -    | 0.9  | us   |
| t <sub>SU_STO</sub> | Set up time for STOP condition                                                              | 0.6    | -    | -    | us   |
| t <sub>BUF</sub>    | Bus free time between a STOP and START condition                                            | 1.3    | -    | -    | us   |
| C <sub>b</sub>      | Capacitive load for each bus line                                                           | -      | -    | 400  | pF   |

**7. Optical Characteristics:**

| Item                              | Symbol | Conditions                               | Specifications |      |     | Unit | Note    |
|-----------------------------------|--------|------------------------------------------|----------------|------|-----|------|---------|
|                                   |        |                                          | Min            | Typ  | Max |      |         |
| Transmittance<br>(with polarizer) | T(%)   | -                                        | 4.2            | 4.8  | -   | -    | -       |
| NTSC                              | %      | $\theta=0$                               | -              | 60   | -   | -    | -       |
| Contrast Ratio                    | CR     | $\theta=0$<br>Normal<br>Viewing<br>angle | 800            | 1000 | -   | -    | (1) (2) |
| Response time                     | TR+TF  | -                                        | -              | 25   | 35  | ms   | (1) (3) |
| Viewing<br>angle                  | Hor.   | $\Theta_{x+}$                            | CR $\geq 10$   | -    | 80  | -    | deg.    |
|                                   |        | $\Theta_{x-}$                            |                | -    | 80  | -    |         |
|                                   | Ver.   | $\Theta_{y+}$                            |                | -    | 80  | -    |         |
|                                   |        | $\Theta_{y-}$                            |                | -    | 80  | -    |         |
|                                   |        |                                          |                | -    | 80  | -    |         |

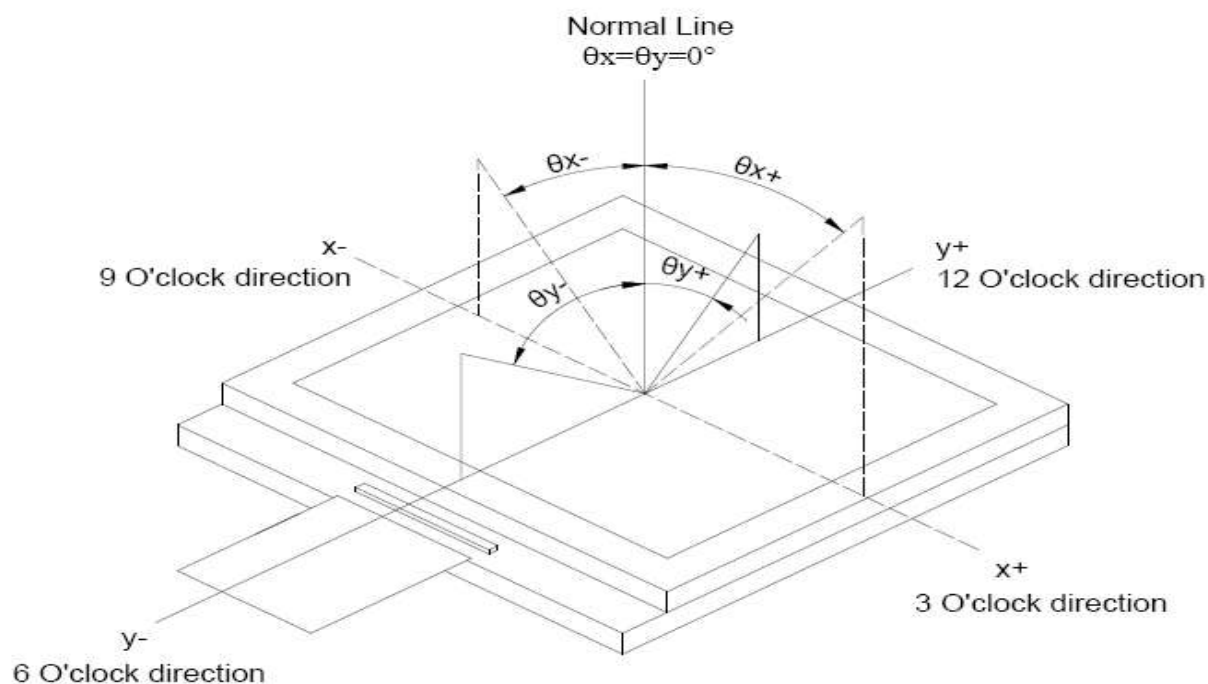
**Measuring Condition**

1. Measuring surrounding: dark room
2. Ambient temperature:  $25 \pm 2^\circ\text{C}$
3. 30 min. Warm-up time.

**Color of CIE Coordinate:**

| Item                                          |       | Symbol | Condition                                                     | Min.   | Typ.   | Max.   |
|-----------------------------------------------|-------|--------|---------------------------------------------------------------|--------|--------|--------|
| Chromaticity<br>Coordinates<br>(Transmissive) | Red   | x      | $\theta = \varphi = 0^\circ$<br>LED Backlight<br>Color Degree | 0.5868 | 0.6368 | 0.6868 |
|                                               |       | y      |                                                               | 0.2853 | 0.3353 | 0.3853 |
|                                               | Green | x      |                                                               | 0.2804 | 0.3304 | 0.3804 |
|                                               |       | y      |                                                               | 0.5327 | 0.5827 | 0.6327 |
|                                               | Blue  | x      |                                                               | 0.0929 | 0.1429 | 0.1929 |
|                                               |       | y      |                                                               | 0.0135 | 0.0635 | 0.1135 |
|                                               | White | x      |                                                               | 0.2627 | 0.3127 | 0.3627 |
|                                               |       | y      |                                                               | 0.2876 | 0.3376 | 0.3876 |

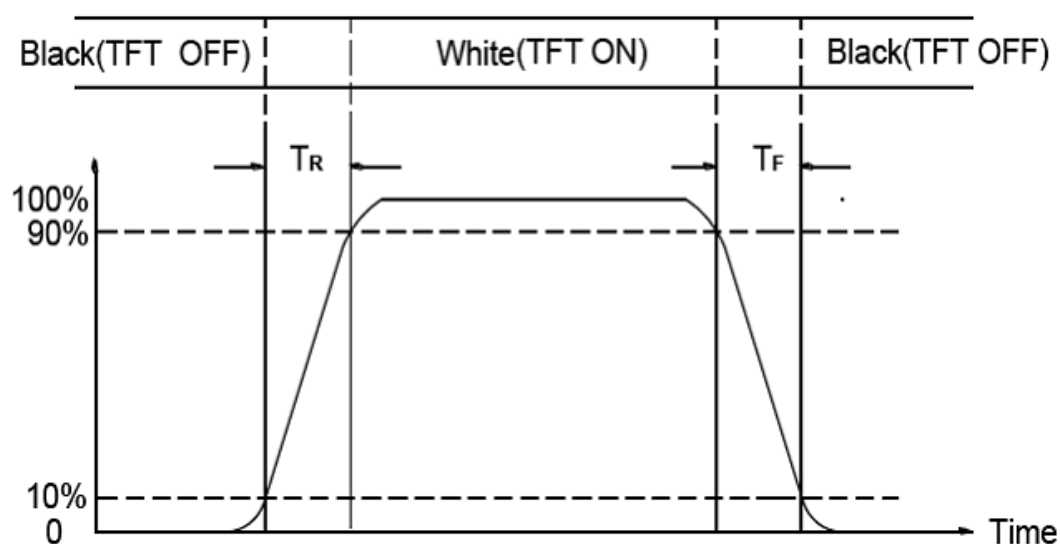
Note (1) Definition of Viewing Angle :



Note (2) Definition of Contrast Ratio(CR) :  
measured at the center point of panel

$$\text{Contrast ratio (CR)} = \frac{\text{Photo detector output when LCD is at "White" state}}{\text{Photo detector output when LCD is at "Black" state}}$$

Note (3) Definition of Response Time : Sum of TR and TF



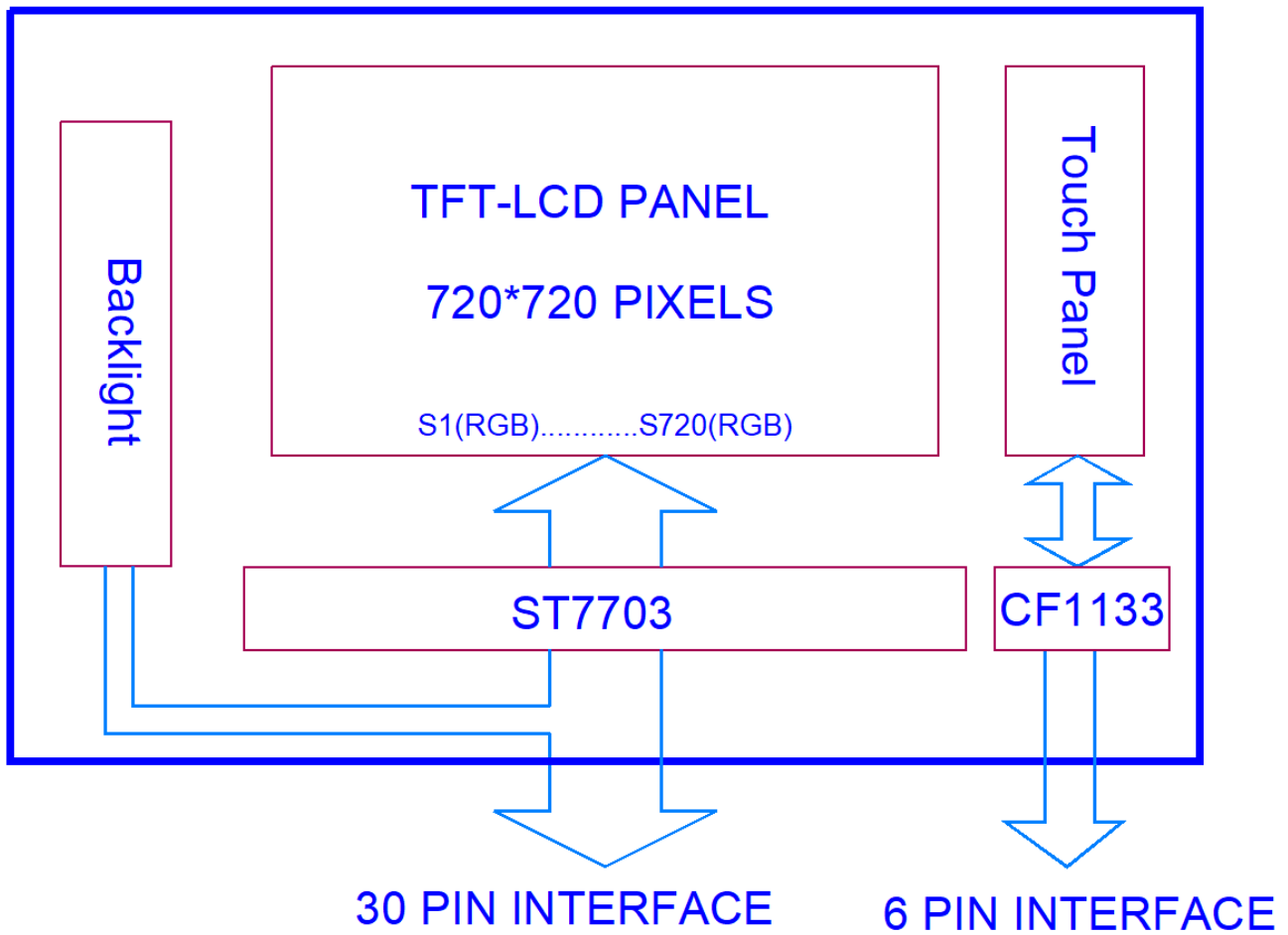
**8. Interface Pin Assignment:****8-1 LCM FPC Interface**

| No. | Symbol | Function                                                        | Remark |
|-----|--------|-----------------------------------------------------------------|--------|
| 1   | LED-A  | LED anode                                                       |        |
| 2   | LED-K  | LED cathode                                                     |        |
| 3   | LED-K  | LED cathode                                                     |        |
| 4   | VCI    | Power Supply                                                    |        |
| 5   | IOVCC  | Power Supply                                                    |        |
| 6   | RESET  | Reset signal input terminal .                                   |        |
| 7   | TE     | Tearing effect                                                  |        |
| 8   | PWM    | This pin is connecting with the external LED driver             |        |
| 9   | GND    | Ground                                                          |        |
| 10  | D0P    | High speed interface data differential signal input/output pins |        |
| 11  | D0N    | High speed interface data differential signal input/output pins |        |
| 12  | GND    | Ground                                                          |        |
| 13  | D1P    | High speed interface clock differential signal input pins       |        |
| 14  | D1N    | High speed interface clock differential signal input pins       |        |
| 15  | GND    | Ground                                                          |        |
| 16  | CLKP   | High speed interface clock differential signal input pins       |        |
| 17  | CLKN   | High speed interface clock differential signal input pins       |        |
| 18  | GND    | Ground                                                          |        |
| 19  | D2P    | High speed interface data differential signal input pins        |        |
| 20  | D2N    | High speed interface data differential signal input pins        |        |
| 21  | GND    | Ground                                                          |        |
| 22  | D3P    | High speed interface data differential signal input pins        |        |
| 23  | D3N    | High speed interface data differential signal input pins        |        |
| 24  | GND    | Ground                                                          |        |
| 25  | NC     | NC                                                              |        |
| 26  | NC     | NC                                                              |        |
| 27  | NC     | NC                                                              |        |
| 28  | NC     | NC                                                              |        |
| 29  | NC     | NC                                                              |        |
| 30  | NC     | NC                                                              |        |

**8-2 CTP FPC Interface**

| No. | Symbol   | Function                     | Remark |
|-----|----------|------------------------------|--------|
| 1   | TP_INT   | Touch panel interrupt output |        |
| 2   | TP_SDA   | Touch panel I2C data         |        |
| 3   | TP_SCL   | Touch panel I2C clock        |        |
| 4   | TP_RESET | Touch panel reset            |        |
| 5   | TP_VCI   | Touch panel power supply     |        |
| 6   | TP_GND   | Touch panel Ground           |        |



**9. Block Diagram:**

## 10. Backlight:

### 1. Standard Lamp Styles (Edge Lighting Type):

The LED chips are distributed over the edge light area of the illumination unit, which gives the less power consumption:

### 2. The Main Advantages of the LED Backlight are as following:

#### 2.1 The brightness of the backlight can simply be adjusted.

By a resistor or a potentiometer.

### 3. Data About LED Backlight:

(Ta=25°)

| PARAMETER                      | Sym.  | Min.  | Typ. | Max. | Unit  | Test Condition | Note |
|--------------------------------|-------|-------|------|------|-------|----------------|------|
| Supply Current                 | I     | 40    |      |      | mA    | V=12.0V        |      |
| Supply Voltage                 | V     | 10.8  | 12.0 | 13.6 | V     | If=40mA        |      |
| Luminous Intensity for LCM+CTP | IV    | 210   | 280  | -    | cd/m2 | If=40mA        | 2    |
| Uniformity for CTP+LCM         | -     | 70    | -    | -    | %     |                | 3    |
| Life Time                      | -     | 20000 | -    | -    | Hr.   |                | 4    |
| Color                          | White |       |      |      |       |                |      |

NOTE:

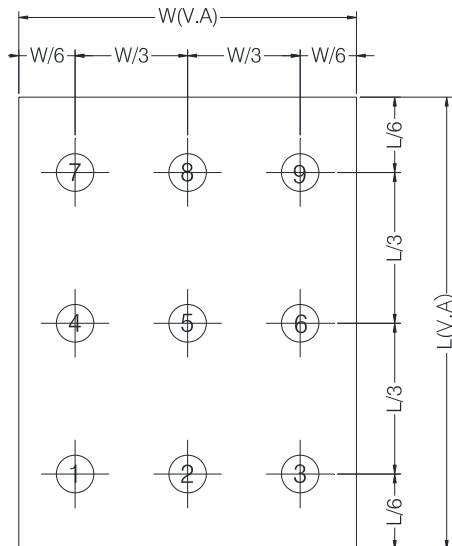
#### 1. Backlight Only

#### 2. Average Luminous Intensity of P1-P9

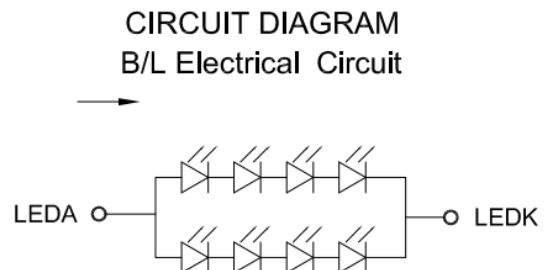
#### 3. Uniformity = Min/Max \* 100%

#### 4. LED life time defined as follow: the final brightness is at 50% of original brightness

### Measured Method: (X\*Y: Light Area)



### Internal Circuit Diagram



### (Effective spatial Distribution)

Using aperture of 1°, distance 50cm

## 11. Standard Specification for Reliability :

### 11-1. Standard Specifications for Reliability

| No | Item                        | Description                                                                                                                                                                                       |
|----|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 | High temperature operation  | The sample should be allowed to stand at 70℃ for 120 hours under driving condition and then returning it to normal temperature condition, and allowing it stand for 2 hours.                      |
| 02 | Low temperature operation   | The sample should be allowed to stand at -20℃ for 120 hours under driving condition and then returning it to normal temperature condition, and allowing it stand for 2 hours.                     |
| 03 | High temperature storage    | The sample should be allowed to stand at 80℃ for 240 hours under no-load condition, and then returning it to normal temperature condition, and allowing it stand for 2 hours.                     |
| 04 | Low temperature storage     | The sample should be allowed to stand at -30℃ for 240 hours under no-load condition, then returning it to normal temperature condition, and allowing it stand for 2 hours.                        |
| 05 | Moisture storage            | The sample should be allowed to stand at 60℃,90%RH MAX for 240 hours under no-load condition, then taking it out and drying it at normal temperature for 2 hours.                                 |
| 06 | Thermal shock storage       | The sample should be allowed to stand the following 10 cycles :<br>-30℃ for 30 minutes → normal temperature for 5 minutes → +80℃ for 30 minutes → normal temperature for 5 minutes, as one cycle. |
| 07 | Packing vibration           | Frequency range : 10Hz ~ 55Hz<br>Amplitude of vibration : 1.5mm      Sweep time: 12 min<br>X,Y,Z 2 hours for each direction.                                                                      |
| 08 | Packing drop test           | According to ISTA 1A 2001.                                                                                                                                                                        |
| 09 | Electrical Static Discharge | Air: ±6KV 150pF/330Ω 5 times                                                                                                                                                                      |
|    |                             | Contact: ±4KV 150pF/330Ω 5 time                                                                                                                                                                   |

\*Sample size for each test item is 3~5pcs

## 11 - 2. Testing Conditions and Inspection Criteria

For the final test the testing sample must be stored at room temperature for 24 hours, after the tests listed in Table 11-1, Standard specifications for Reliability have been executed in order to ensure stability.

| No | Item                | Test Model             | In section Criteria                                                                                                |
|----|---------------------|------------------------|--------------------------------------------------------------------------------------------------------------------|
| 01 | Current Consumption | Refer To Specification | The current consumption should conform to the product specification.                                               |
| 02 | Contrast            | Refer To Specification | After the tests have been executed, the contrast must be larger than half of its initial value prior to the tests. |
| 03 | Appearance          | Visual inspection      | Defect free.                                                                                                       |

## 11- 3. MTBF

|      |                                                                                                                                                                                                                                                                                           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MTBF | Functions, performance, appearance, etc. shall be free from remarkable deterioration within 50,000 hours under ordinary operating and storage conditions room temperature ( $25\pm 5^{\circ}\text{C}$ ), normal humidity ( $50\pm 10\%$ RH), and in area not exposed to direct sun light. |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## **12. Specification of Quality Assurance:**

### 12-1. Purpose

This standard for Quality Assurance should affirm the quality of LCD module products to supply to purchaser by YEEBO CORPORATION (Supplier).

### 12-2. Standard for Quality Test

#### a. Inspection:

Before delivering, the supplier should take the following tests, and affirm the quality of product.

#### b. Electro-Optical Characteristics:

According to the individual specification to test the product.

#### c. Test of Appearance Characteristics:

According to the individual specification to test the product.

#### d. Test of Reliability Characteristics:

According to the definition of reliability on the specification for testing products.

#### e. Delivery Test:

Before delivering, the supplier should take the delivery test.

(i) Test method: According to **ISO2859-1**. General Inspection Level II take a single time.

(ii) The defects classify of AQL as following:

Major defect: AQL = 0.65

Minor defect: AQL = 2.5

Total defects: AQL = 2.5

### 12-3. Non- conforming Analysis & Deal With Manners

#### a. Non- conforming Analysis:

(i) Purchaser should supply the detail data of non- conforming sample and the non-conforming.

(ii) After accepting the detail data from purchaser, the analysis of non- conforming should be finished in two weeks.

(iii) If supplier can not finish analysis on time, must announce purchaser before 3 days.

#### b. Disposition of non- conforming:

(i) If find any product defect of supplier during assembly time, supplier must change the good product for every defect after recognition.

(ii) Both supplier and customer should analyze the reason and discuss the disposition of non- conforming when the reason of nonconforming is not sure.

### 12-4. Agreement items

Both sides should discuss together when the following problems happen.

a. There is any problem of standard of quality assurance, and both sides should think that must be modified.

b. There is any argument item which does not record in the standard of quality assurance.

c. Any other special problem.

## 12-5. Standard of The Product Appearance Test

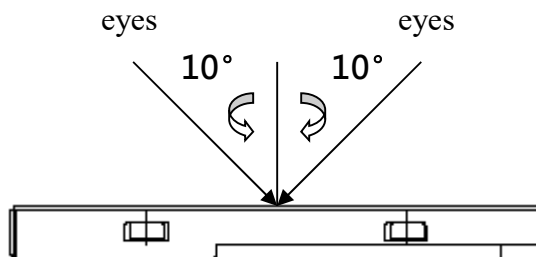
## a. Manner of appearance test:

(i) The test must be under  $20W \times 2$  or  $40W$  fluorescent light, and the distance of view must be at  $30 \pm 5\text{cm}$ .

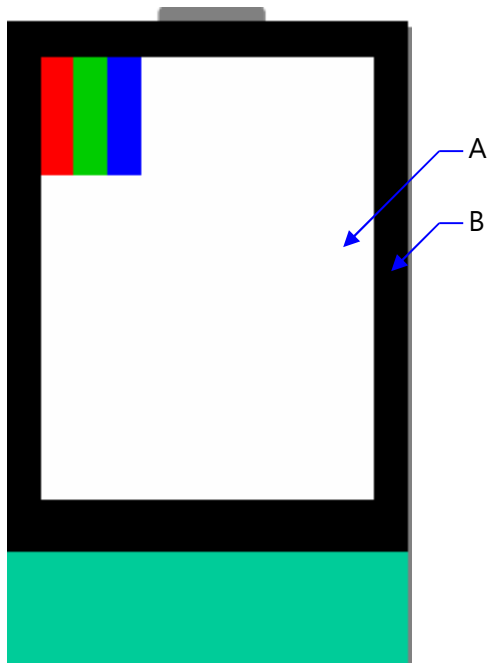
(ii) When test the model of transmissive product must add the reflective plate.

(iii) The test direction is base on around  $10^\circ$  of vertical line.

(iii) Temperature:  $25 \pm 5^\circ\text{C}$  Humidity:  $60 \pm 10\%\text{RH}$



## (iv) Definition of area:



A. Area: Viewing area.

B. Area: Out of viewing area.  
(Outside viewing area)

## b. Basic principle:

(i) It will accord to the AQL when the standard can not be described.

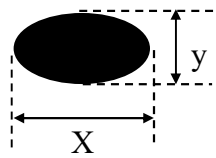
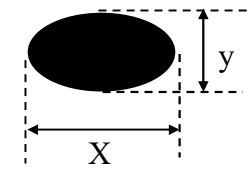
(ii) The sample of the lowest acceptable quality level must be discussed by both supplier and customer when any dispute happened.

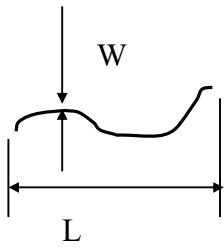
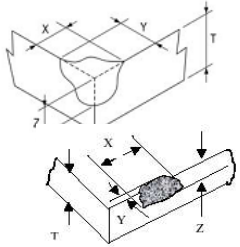
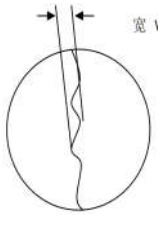
(iii) Must add new item on time when it is necessary.

## c. Standard of inspection: (Unit: mm)

## 12-6. Inspection specification

Defect out of viewing area can be neglected.

| Item                                                                                            | Specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Unit : mm | AQL                |             |                                              |                      |   |                      |   |            |    |  |     |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------|-------------|----------------------------------------------|----------------------|---|----------------------|---|------------|----|--|-----|
| Electrical Testing                                                                              | 1.1 Open<br>1.2 Short<br>1.3 T/P failure<br>1.4 Missing vertical, horizontal segment, segment contrast defect.<br>1.5 Missing character, dot or icon.<br>1.6 Display malfunction.<br>1.7 No function or no display.<br>1.8 Current consumption exceeds product specifications.<br>1.9 LCD viewing angle defect.<br>1.10 Mixed product types.<br>1.11 Flicker                                                                                                                                                                                                                                                                                                                                                                                                                                                            |           | 0.65               |             |                                              |                      |   |                      |   |            |    |  |     |
| explosion-proof film<br>bubble/Concave and convex point/indentation / Contamination             | <table><tr><th>D</th><th>Acceptable numbers</th></tr><tr><td><math>\leq 0.2</math></td><td>ignored (No more than five spots within 5mm)</td></tr><tr><td><math>0.2 &lt; D \leq 0.5</math></td><td>3</td></tr><tr><td><math>0.5 &lt; D \leq 1.0</math></td><td>2</td></tr><tr><td><math>D &gt; 1.0</math></td><td>NG</td></tr></table> <div><p><math>D = (x+y) / 2</math></p></div> <div><div>1. Product's front side checked according to this specification, back side ignored, but light leakage is not allowed.</div><div>2. Printing ink peel off is not allowed.</div><div>3. The particle will be ignored when it is removable by cleaning</div></div> <div>* Densely spaced: No more than two spots within 10mm</div>         | D         | Acceptable numbers | $\leq 0.2$  | ignored (No more than five spots within 5mm) | $0.2 < D \leq 0.5$   | 3 | $0.5 < D \leq 1.0$   | 2 | $D > 1.0$  | NG |  | 2.5 |
| D                                                                                               | Acceptable numbers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |                    |             |                                              |                      |   |                      |   |            |    |  |     |
| $\leq 0.2$                                                                                      | ignored (No more than five spots within 5mm)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |           |                    |             |                                              |                      |   |                      |   |            |    |  |     |
| $0.2 < D \leq 0.5$                                                                              | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |                    |             |                                              |                      |   |                      |   |            |    |  |     |
| $0.5 < D \leq 1.0$                                                                              | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |                    |             |                                              |                      |   |                      |   |            |    |  |     |
| $D > 1.0$                                                                                       | NG                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |                    |             |                                              |                      |   |                      |   |            |    |  |     |
| Black spots /<br>White spots<br>/Bright spots/<br>Color spots<br>/polluted inside/<br>punctured | <table><tr><th>D</th><th>Acceptable numbers</th></tr><tr><td><math>\leq 0.15</math></td><td>ignored (No more than five spots within 5mm)</td></tr><tr><td><math>0.15 &lt; D \leq 0.25</math></td><td>3</td></tr><tr><td><math>0.25 &lt; D \leq 0.35</math></td><td>2</td></tr><tr><td><math>D &gt; 0.35</math></td><td>NG</td></tr></table> <div><p><math>D = (x+y) / 2</math></p></div> <div><div>1. Product's front side checked according to this specification, back side ignored, but light leakage is not allowed.</div><div>2. Printing ink peel off is not allowed.</div><div>3. The particle will be ignored when it is removable by cleaning</div></div> <div>* Densely spaced: No more than two spots within 10mm</div> | D         | Acceptable numbers | $\leq 0.15$ | ignored (No more than five spots within 5mm) | $0.15 < D \leq 0.25$ | 3 | $0.25 < D \leq 0.35$ | 2 | $D > 0.35$ | NG |  | 2.5 |
| D                                                                                               | Acceptable numbers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |                    |             |                                              |                      |   |                      |   |            |    |  |     |
| $\leq 0.15$                                                                                     | ignored (No more than five spots within 5mm)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |           |                    |             |                                              |                      |   |                      |   |            |    |  |     |
| $0.15 < D \leq 0.25$                                                                            | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |                    |             |                                              |                      |   |                      |   |            |    |  |     |
| $0.25 < D \leq 0.35$                                                                            | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |                    |             |                                              |                      |   |                      |   |            |    |  |     |
| $D > 0.35$                                                                                      | NG                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |                    |             |                                              |                      |   |                      |   |            |    |  |     |

|                                                                                                                         |                                                                                                                                                                                                                                                                                                                        |  |          |                                              |                                                                                                                         |      |              |                    |                                                      |                                                                                   |                                                                                       |     |
|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------|--------------|--------------------|------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----|
| Linear Object:<br>Fiber, scurf,<br>scratches and other<br>linear defects (not<br>affecting function)                    | W                                                                                                                                                                                                                                                                                                                      |  | L        | Acceptable numbers                           |                                      | 2.5  |              |                    |                                                      |                                                                                   |                                                                                       |     |
|                                                                                                                         | $\leq 0.05$                                                                                                                                                                                                                                                                                                            |  | $\leq 5$ | ignored (No more than five lines within 5mm) |                                                                                                                         |      |              |                    |                                                      |                                                                                   |                                                                                       |     |
|                                                                                                                         | $0.05 < W \leq 0.15$                                                                                                                                                                                                                                                                                                   |  | $\leq 5$ | 2                                            |                                                                                                                         |      |              |                    |                                                      |                                                                                   |                                                                                       |     |
|                                                                                                                         | $W > 0.15$                                                                                                                                                                                                                                                                                                             |  |          | NG                                           |                                                                                                                         |      |              |                    |                                                      |                                                                                   |                                                                                       |     |
|                                                                                                                         | The reverse side scratches, not affect to the electronic circuit, cannot find the scratches from the front side is acceptable                                                                                                                                                                                          |  |          |                                              |                                                                                                                         |      |              |                    |                                                      |                                                                                   |                                                                                       |     |
| * Densely spaced: No more than two lines within 10mm                                                                    |                                                                                                                                                                                                                                                                                                                        |  |          |                                              |                                                                                                                         |      |              |                    |                                                      |                                                                                   |                                                                                       |     |
| Glass edge chipping、edge breakage                                                                                       | <table><tr><td colspan="2">Edge breakage can't affect visual effect (edge breakage can't cause damage to circuit); over lens have no visual damage</td></tr><tr><td>conditions</td><td>Acceptable numbers</td></tr><tr><td><math>X \leq 1.0\text{mm}, Y \leq 1.0\text{mm}, Z \leq T</math></td><td>2</td></tr></table> |  |          |                                              | Edge breakage can't affect visual effect (edge breakage can't cause damage to circuit); over lens have no visual damage |      | conditions   | Acceptable numbers | $X \leq 1.0\text{mm}, Y \leq 1.0\text{mm}, Z \leq T$ | 2                                                                                 |   | 2.5 |
| Edge breakage can't affect visual effect (edge breakage can't cause damage to circuit); over lens have no visual damage |                                                                                                                                                                                                                                                                                                                        |  |          |                                              |                                                                                                                         |      |              |                    |                                                      |                                                                                   |                                                                                       |     |
| conditions                                                                                                              | Acceptable numbers                                                                                                                                                                                                                                                                                                     |  |          |                                              |                                                                                                                         |      |              |                    |                                                      |                                                                                   |                                                                                       |     |
| $X \leq 1.0\text{mm}, Y \leq 1.0\text{mm}, Z \leq T$                                                                    | 2                                                                                                                                                                                                                                                                                                                      |  |          |                                              |                                                                                                                         |      |              |                    |                                                      |                                                                                   |                                                                                       |     |
| Glass broken                                                                                                            | Visual broken is NG, and there is no potential fault.                                                                                                                                                                                                                                                                  |  |          |                                              |                                                                                                                         | 0.65 |              |                    |                                                      |                                                                                   |                                                                                       |     |
| 1. V/A printed edges sawtooth inspected according to this standard<br>2. LOGO's sawtooth                                | <table><tr><td colspan="2">Some contentious defect judged according to samples</td></tr><tr><td>Product type</td><td>Conditions</td></tr><tr><td>Same size</td><td>1、width below 0.2 inch (included) ignored, above 0.2 NG<br/>2、Length not accounted</td></tr></table>                                                |  |          |                                              | Some contentious defect judged according to samples                                                                     |      | Product type | Conditions         | Same size                                            | 1、width below 0.2 inch (included) ignored, above 0.2 NG<br>2、Length not accounted |  | 2.5 |
| Some contentious defect judged according to samples                                                                     |                                                                                                                                                                                                                                                                                                                        |  |          |                                              |                                                                                                                         |      |              |                    |                                                      |                                                                                   |                                                                                       |     |
| Product type                                                                                                            | Conditions                                                                                                                                                                                                                                                                                                             |  |          |                                              |                                                                                                                         |      |              |                    |                                                      |                                                                                   |                                                                                       |     |
| Same size                                                                                                               | 1、width below 0.2 inch (included) ignored, above 0.2 NG<br>2、Length not accounted                                                                                                                                                                                                                                      |  |          |                                              |                                                                                                                         |      |              |                    |                                                      |                                                                                   |                                                                                       |     |
| Specific dimension                                                                                                      | In accordance with product outline drawing or specification (key dimension) or engineering sample.                                                                                                                                                                                                                     |  |          |                                              |                                                                                                                         | 2.5  |              |                    |                                                      |                                                                                   |                                                                                       |     |
| Glue overflow/Frame                                                                                                     | 1. Glue overflow exceed 0.2mm to the black frame is not allowed.                                                                                                                                                                                                                                                       |  |          |                                              |                                                                                                                         | 2.5  |              |                    |                                                      |                                                                                   |                                                                                       |     |



|     |                                 |                                                                                                                                                                                                                      |      |
|-----|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| FPC | Bonding bubble/<br>Misalignment | FPC golden finger hot pressure's bubble or impurity diameter shall be below 1/2 of the pressed area, pressed deviation shall not exceed 1/2 of the silver line width, and 40X microscope cannot have obvious cracks. | 0.65 |
|     | Folded mark<br>(minor fault)    | Linearity irreversibility folded mark and acute angle folded mark is NG.                                                                                                                                             | 2.5  |
|     | EMI FILM<br>(minor fault)       | Surface broken, scratched $\leq 0.3\text{mm}$<br>Surface broken below 5mm can be modified by print ink, after modified, the result shall be achieved to EMI                                                          | 2.5  |

## **13. Handling Precaution:**

### **13.1 Warranty**

This product has been manufactured to specifications as a part for use in your company's general electronic products. It is guaranteed to perform according to delivery specifications. For any other use apart from general electronic equipment, we will not take responsibility if the product is used in medical devices, nuclear power control equipment, aerospace equipment, fire and security systems, or any other applications in which there is a direct risk to human life and where extremely high levels of reliability are required. If the product is to be used in any of the above applications, we will need to enter into a separate product liability agreement.

1. We cannot accept responsibility for any defect arise after additional process of the product (including disassembly and reassembly), after product delivery.
2. We cannot accept responsibility for any defect, which may arise after the application of strong external force to the product.
3. We cannot accept responsibility for any defect, which may arise due to the application of static electricity after the product has passed your company's acceptance inspection procedures.
4. We cannot accept responsibility for industrial property, which may arise through the use of your product, with exception to those issues relating directly to the structure or method of manufacturing of our product 3months from YEEBO production.
5. The liability of YB is limited to repair or replacement on the terms set forth below. YB will not be responsible for any subsequent or consequential events or injury or damage to any personnel or user including third party personnel and/or user. Unless otherwise agreed in writing between YB and the customer, YB will only replace or repair any of its CTP which is found defective electrically or visually when inspected in accordance with YB GENERAL CTP INSPECTION STANDARD.

### **13.2. Precautions in Use of CTP Module**

#### **13.2-1. Handling of CTP Module**

13.2-1-1 Please operate the capacitive touch panel by touch the panel surface with finger or electric pen

13.2-1-2 Store the products at the temperature and humidity mentioned in the specification in a good package do not expose the products under direct sunlight.

13.2-1-3 Do not hit the capacitive touch panel in strong force , or drop it down, it is made of glass and friable.

13.2-1-4 Put on finger coats · gloves or mask to protect the products from fingerprint of stain. Do not upload/unload the touch panel by holding the FPC cable. Do not bend the FPC cable often or pull it hard when installing, as FPC cable is soft and connected to touch panel body.

13.2-1-5 Pay attention to the prevention from high voltage and static electricity.

#### **13.2-2 Storage**

13.2-2-1 Store in ambient temperature of  $25\pm 5^{\circ}\text{C}$  , and relative humidity of  $50\pm 10\%\text{RH}$ . Do not expose to sunlight or fluorescent light.

13.2-2-2 Storage in a clean environment, free from dust, active gas, and solvent.

13.2-2-3 Store in anti-static electricity container.

13.2-2-4 Store without any physical load.

13.2-2-5 Appearance, 3months; Function, 1 year; within the validity, failed CTP can be replaced 1 to 1

### **13.3 Guarantee**

Our products meet requirements of the environment. YEEBO ROHS requirement is based on European Union Directive 2011/65/EU (ROHS) Requirements and Update.